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RESEARCH ARTICLE

Design of an Iterative Model for Incremental Enhancements in Quantum Image Processing Using Reinforcement Learning-Based Optimizations

LALITHA KUMARI PAPPALA[1], SAI BABU VEESAM[1], KANNAIAH CHATTU[1], JANAPATI VENKATA KRISHNA[2], JYOSTNA DEVI BODAPATI[3], (Senior Member, IEEE), AND B. TARAKESWARA RAO[4]

[1]School of Computer Science and Engineering, VIT-AP University, Amaravati, Andhra Pradesh 522237, India
[2]Unit Head of IoT and Robotics and AI and AR&VR Departments, Institute of Engineering and Technology, Srinivas University, Mangaluru 574146, India
[3]Department of Advanced Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research, Vadlamudi, Andhra Pradesh 522213, India
[4]Department of CSE, Kallam Haranadhareddy Institute of Technology, Guntur, Andhra Pradesh 522019, India

Corresponding author: Lalitha Kumari Pappala (lalitha.p@vitap.ac.in)

ABSTRACT In this text we propose a collection of enhanced quantum methods that would drastically enhance the efficiency and accuracy of biomedical image processing. There are four key methodologies that have been proposed in this work: Quantum Support Vector Machines with Enhanced Feature Extraction, Quantum Generative Adversarial Networks for Data Augmentation, Quantum Entanglement and Correlation Analysis for Feature Detection, and Quantum-Enhanced Image Reconstruction for Noisy Data Samples. Q-SVM-EFE is a Quantum Support Vector Machine-based framework that incorporates advanced feature extraction to map classical image data into high-dimensional quantum states, which should capture intricate patterns and therefore be more accurate for classification. This approach scored more than 95 percent in terms of classification accuracy and tenfold increased the speed of training compared to classical SVMs. QGAN-DA generates high-quality synthetic images to alleviate the scarcity of labeled medical data. It trains a generator network through quantum circuits for the generation of new samples and a discriminator network for real image and sample classification, thus ensuring high diversity and quality in synthetic data with SSIM values above 0.90. QECA-FD applies quantum entanglement in the analysis of biomedical image correlations and is therefore very effective in detecting subtle features and anomalies. This resulted in over 95% sensitivity in anomaly detection and increased the speed of the correlation analysis by five times. QIR-ND enables enhanced image reconstruction from sullied or incomplete data through quantum noise reduction and reconstruction techniques. This new approach gave a PSNR greater than 40 dB and an eightfold improvement in reconstruction delays.

INDEX TERMS Biomedical imaging, generative adversarial networks, image reconstruction, quantum computing, support vector machines.

I. INTRODUCTION

The new field of quantum computing holds bright promises for advances in many areas of science and engineering, one of which is biomedical image processing. Quantum algorithms can make a big difference in this area due to their computational power. Conventional biomedical image processing techniques are generally limited in success by the enormous quantity and complexity of data they need to process in the procedure [1], [2], [3]. Finally, high

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dimensional biomedical images like MRI and CT scans need robust methods to accurately classify, detect, and reconstruct the critical features. Classical methods of image processing, at times, are not powerful enough to process and meet up to the efficiency of algorithms. Image processing using the classical approach normally involves sophisticated algorithms in feature extraction, classification, and noise reduction. These methods, however, have a dimensionality problem for high-dimensional data and are mainly restricted to needing gigantic portions of labeled training data for high accuracy. Furthermore, the intrinsic noise and incomplete data in the majority of biomedical images may cause suboptimal reconstruction and analysis. All of this creates an urge to look into alternative approaches that could better exploit the complex patterns and correlations in biomedical data. Quantum computing provides solutions to such challenges by allowing fundamentally different ways of processing and analyzing data. Such principles as superposition and entanglement allow quantum algorithms to execute some computations up to an exponentially faster degree than the classical ones. Such speedup is especially appropriate for tasks that involve high-dimensional data, and therefore, quantum computing would be an ideal candidate to enhance biomedical image processing techniques. It is in these regards that this work presents a quantum-enhanced package of methods to improve efficiency and accuracy in biomedical image processing. In this set will be Quantum Support Vector Machines with Enhanced Feature Extraction, Quantum Generative Adversarial Networks for Data Augmentation, Quantum Entanglement and Correlation Analysis for Feature Detection, and Quantum Image Reconstruction for Noisy Data. Every method is specialized to meet the challenges, in general, during biomedical imaging, and exploits the unique capabilities brought by quantum computing to return better results than one might hope from a naive application of current computer techniques.

In the recent past, quantum innovation has emerged as a critical force in biomedical image processing to accomplish dramatic potentiality that may enhance the accuracy, efficiency, and reliability in medical diagnostics. Biomedical images, such as MRI, CT, and PET scans, consist of high-dimensional data which prove tricky for traditional techniques of image processing, especially when the data is noisy, of sparse datasets, or subtle anomalies present. Such accurate processing of images will only be possible for early disease detection, treatment planning, and research. This quantum computing will introduce new revolutionary techniques for dealing with data through quantum superposition, entanglement, and parallelism. Such capabilities in computation open up the use cases for the quantum-enhanced algorithms and address the paramount issues of biomedical imaging, such as large-scale datasets and weakly patterned signals, faster computations that would be slower than manageable for classical systems. Hence, quantum innovation is of extreme importance to a revolution in medical image processing that will both affect the accuracy and speed of diagnostic tools. With recent developments in quantum technologies, quantum computing is being brought one step closer to practical realization, especially in areas like quantum machine learning, quantum neural networks, and enhancement of optimization techniques through quantum methods. Quantum algorithm developments in applications such as image segmentation, image classification, data augmentation, and noise reduction appear to promise significant efficiency enhancements for further promotion of biomedical image analysis performance. For instance, it is now that quantum support vector machines (SVMs) will replace classical models because they now demonstrate better accuracy and speed of performance, whereas quantum generative adversarial networks (QGANs) can now learn to generate high-quality synthetic data that will replace limited medical datasets with appropriately abundant data to complement the datasets. On top of that, integrating quantum entanglement for correlation analysis has opened new avenues for detection of subtle anomalies in medical images that would otherwise pass undetected. This would see increased usage of quantum-enhanced techniques in the biomedical image processing due to the growing improvement of quantum hardware concerning qubit fidelity, error rates, and scalability, which would make the systems more accurate and efficient in medical diagnostics.

The management of crowds in public spaces and massive events has turned out to be highly indispensable due to the increasing complexity of urban environments, mass gatherings, and frequency of crowds. Crowds have been managed based on static models and heuristic rules and are incapable of addressing dynamic crowd behavior that can often be unpredictable. The inefficiencies also turn out to be considerable safety risks in high-density situations. This allows for the use of advanced computational techniques such as machine learning, computer vision, and predictive analytics in real-time with the capacity to adapt the situation to increase predictive precision and optimize resources. Crowd analysis has also just recently shown the ability how different methodologies can be effectively combined to analyze and manage crowds. For example, the Quantum K-nearest Neighbor Classification Algorithm [1] has demonstrated the possibility of employing quantum-enhanced algorithms toward large-scale data classification as well as decision-making with applicability to crowd management. Another example is about agent-based modeling, emerging as a powerful tool simulating individual and collective behaviors of crowds and especially when merged with reinforcement learning for optimizing decisions dynamically by agents.

Convolutional neural networks, deep learning have proved highly instrumental in crowd density estimation and real-time alerting systems. Recently, with works such as segmentation-guided attention networks [2] and transformer-based architectures for feature aggregation [3], it is seen that they achieve an impressive accuracy for crowd counting and localization tasks. Still, these methods usually require massive amounts of labeled datasets and heavy

computational resources, which reduces their scalability in real-time scenarios. Predictive analytics techniques such as ARIMA and Long Short-Term Memory (LSTM) networks have been quite promising in the crowd density and transportation demand forecasts. The time-series-based forecasting models are particularly useful in proactive resource allocations and operational planning but appear to have certain limitations wherein non-stationarity might be present, or possibly sparse data is encountered.

The Internet of Things and edge computing technologies have also played a very significant role in this field where the data is collected and processed in real-time. The use of IoT-based frameworks for crowd monitoring [4] and privacy-preserving crowd analysis using Wi-Fi sensing [5] have successfully proven the feasibility of the decentralized systems in the context of a smart city environment. These methods, however, have the limitation of being dependent on infrastructure and quality streams. Multi-agent pathfinding algorithms with heuristic optimization and reinforcement learning also have been exploited in optimizing individual movements within shared spaces, reducing congestions, and increasing efficiency in flow [6].

The paper addresses the present approach's weakness and attempted to propose an integrated framework combining ABM along with reinforcement learning, CNN in a real-time crowd density estimate, predictive analytics based on ARIMA and LSTM, and dynamic path optimization by application of MAPF. Contrasting with the proposed framework, it focuses on diverse and high-density crowd scenarios, hence handling them all holistically towards a solution for real-time crowd management. There has been an improvement in movement efficiency up to 28.5%, 32.1% of congestion reduction, and also a crowd density prediction with the precision rate of 88.7%, which is higher compared to the state-of-art methods.

This paper shows great strides in the integration of quantum computing principles into biomedical image processing to solve some of the long-standing problems in this field. The quantum algorithms changed the game through the unique features such as superposition, entanglement, and quantum parallelism, thereby making it possible to solve problems that could not be solved classically. For instance, Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE) illustrates how quantum kernels can map classical image data into high-dimensional quantum states, preserving intricate features and patterns that are otherwise missed by classical methods. This has resulted in classification accuracies of over 96.4%, a tenfold reduction in computational time, and an improvement of several orders of magnitude over traditional methods. Similarly, Quantum Generative Adversarial Networks for Data Augmentation, QGAN-DA uses quantum circuits to produce synthetic biomedical images that have SSIM values higher than 0.93 and which are overcome by data scarcity, hence the application of deep learning techniques has been limited in the process of medical imaging process.

Another improvement is quantum features that involve an evolution of the methods with principles redefining the traditional methods for feature detection and image reconstruction. QECA-FD based on quantum entanglement analyzes subtle anomalies at 97% sensitivity levels but takes five times longer to accomplish classical correlation analysis. Furthermore, QIR-ND applies quantum noise reduction and reconstruction techniques toward peak signal-to-noise ratios (PSNR) higher than 42.0 dB while reducing reconstruction time down to 3.2 seconds, much better than the competing techniques. Altogether, the two approaches in quantum enhancement represent a tremendous leap forward in biomedical image processing, not only proving how the principles of quantum computing can address these very problems but also showing potential breakthroughs in increasing the accuracy of diagnostics, as well as efficiency and reliability in medical imaging applications.

Q-SVM-EFE integrates state-of-the-art feature extraction into quantum support vector machines. In this methodology, quantum kernels are used to map classical image data into high-dimensional quantum states while retaining relevant features and patterns that capture information necessary for correct classification. A quantum SVM is trained on this quantum-encoded data to optimize the hyperplane effectively for classification. Preliminary results indicate that Q-SVM-EFE can go beyond 95% in classification accuracy for classical biomedical image datasets, with the additional benefit of training speedup by a factor of ten over classical SVMs. QGAN-DA solves the challenge of limited labeled medical data by generating high-quality synthetic biomedical images. A quantum generator network is used to create new image samples from quantum noise. A quantum-enhanced discriminator network distinguishes between real and synthetic images. It makes the generator very strong, and hence these synthetic images ensure high diversity with realistic medical features. QGAN-DA-created synthetic images have been tested to get SSIM values higher than 0.90 when compared against real images, an indication of quality in the synthetic images. QECA-FD explores quantum entanglement in the analysis of correlations within biomedical images and offers better detection of subtle features and anomalies. Due to entanglement, different regions of the image are computationally able to compute quantum correlations that give way to important patterns and features. The sensitivity in anomaly detection was more than 95%, and processing speed increased five-fold against the classical correlation analysis method. QIR-ND focuses on improving reconstruction quality of biomedical images from noisy or incomplete data samples. By applying quantum algorithms for noise reduction, QIR-ND applies quantum superposition and entanglement to efficiently denoise input data. Subsequently, full-image reconstruction with quantum-enhanced techniques maximizes clarity and accuracy. With regard to this, QIR-ND has obtained reconstructed images whose PSNR is higher than 40 dB, with reconstruction times that are eightfold faster than classical methods. The introduction

of these quantum-enhanced methods thus marks huge steps forward made in biomedical image processing. With the potential brought about by the computational prowess of quantum computing, these new techniques have noticeably improved accuracy, speed, and overall performance. This not only works to illustrate the potential for quantum computing to transform biomedical imaging but lays the groundwork for further research and development in this promising area for this process.

II. MOTIVATION AND CONTRIBUTION

This research is motivated by the urge to break limitations pertaining to the classical biomedical image processing techniques. Biomedical images, and particularly those gotten from modalities such as MRI and CT scans, carry vast volumes of complex, high-dimensional data, which warrant sophisticated analysis for an accurate diagnosis and planning of treatment. Traditional methodologies lack efficiency in processing this data due to high dimensionality and limited numbers of labeled datasets; in fact, most of the data samples are found noisy or incomplete. This makes it impossible for medical professionals to make timely and accurate diagnoses, which stresses the more advanced processing techniques needed. It uses quantum computing ideas in a very new way to obtain a number of vital contributions for biomedical image processing: Primarily, the advancement of classification accuracy and speed was demonstrated by Q-SVM-EFE with the inclusion of advanced extraction technique in feature curves together with quantum support vector machines. More importantly, QGAM-SVM showed high accuracy in capturing the complex patterns and characteristics of high-dimensional biomedical images, rendering a classification accuracy exceeding 95%, yet with ten times speeds in training, when compared with the classical SVM model. Second, the development of QGAN-DA deals with another urgent issue: the slightly dominant number of labels toward the medical data samples. This model guarantees a diversified and realistic dataset for training deep models by generating high-quality synthetic images. The synthetic images generated by QGAN-DA are both high in quality and useful, with a quality and applicability way above 0.90 in SSIM, again, useful for augmenting other datasets or samples. The quantum entanglement in the QECA-FD improved features by detecting biomedical images. It can look for a quantum correlation approach to detect subtle anomalies with high sensitivity such that it detects more than 95% detection sensitivity and is five times faster.

The motivation of the manuscript is to address several critical limitations in classical methods of biomedical image processing that are associated with high-dimensional data, fewer labeled samples, and noise. These four proposed methods, Q-SVM-EFE, QGAN-DA, QECA-FD, and QIR-ND, are deeply connected and constitute an integrated framework intended to tackle a set of problems that, while independent, are nonetheless related within the biomedical image processing pipeline. The approach taken by Q-SVM-EFE enhances the classification accuracy and computational efficiency by taking advantage of the quantum enhancement of feature extraction. The former supports that using synthetic high-quality data helps mitigate the shortfalls from low labeled sample sizes. QECA-FD relies on quantum entanglement correlation for an enhanced sensitivity towards subtle feature anomaly detections. While, QIR-ND strives to enhance the quality and fidelity in recovering image data from noisy or degraded sources. Together, they shape up into a comprehensive solution driven by quantum, hence highly enhanced accuracy, efficiency, and robustness for biomedical image processing that both solve specific and systemic issues at one stroke in process.

This manuscript presents four quantum-enhanced methods: Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE), Quantum Generative Adversarial Networks for Data Augmentation (QGAN-DA), Quantum Entanglement and Correlation Analysis for Feature Detection (QECA-FD), and Quantum-Enhanced Image Reconstruction for Noisy Data (QIR-ND) Process. Thesee motivated contributions addressing some of the more severe bottlenecks and barriers found within biomedical image processing. Q-SVM-EFE increases classification accuracy to over 95% with tenfold faster training times by using quantum kernels for high-dimensional feature mapping. QGAN-DA synthesizes images with SSIM scores higher than 0.90; overcomes data scarcity and enhances the model robustness. QECA-FD detects anomalies with sensitivity values more than 95% by leveraging quantum entanglement to ensure highly precise correlation analysis, and feature detection is five times faster. QIR-ND reconstructs noisy or incomplete images with PSNR values larger than 40 dB, with reconstruction time eightfold faster. These contributions collectively form a comprehensive framework that enhances diagnostic accuracy, data augmentation, feature sensitivity, and reconstruction efficiency of images, hence significantly enhancing the capabilities of biomedical imaging systems.

Finally, QIR-ND significantly enhances the reconstruction quality for biomedical images from noisy or incomplete data samples. Going a step further than this, with state-of-the-art quantum noise reduction and further developed reconstruction methods, QIR-ND obtains reconstructions that offer PSNR values of over 40 dB and reconstructs an image eight times faster. All these contributions come together to provide conclusive exemplification for quantum computing in shaping a new trajectory for biomedical image processing at the service of more accurate, efficient, and credible diagnostic tools. This work highlights not only the most critical issues in the field but also new ways toward research and development, which would consequently improve healthcare in practice.

The proposed framework integrates four advanced quantum-enhanced methodologies for addressing critical challenges in biomedical image processing. The Q-SVM-EFE method maps the classical biomedical image data into the high-dimensional quantum states using the quantum

kernels. This maintains the complex patterns for better accuracy in classification. It is ensured that more than 95% accuracy is obtained while reducing the training time significantly by utilizing quantum parallelism. QGAN-DA is a method of addressing the problem of the limited availability of labeled medical data. It generates synthetic images through quantum circuits for data augmentation. It makes realistic images synthesized from quantum noise, and the discriminator network facilitates the training process by allowing for a structural similarity index more than 0.90 to be achieved in synthetic images, thus complementing training datasets for improved model performance.

Quantum Entanglement and Correlation Analysis for Feature Detection (QECA-FD) is designed with the use of quantum entanglement that facilitates detecting tiny features or anomalies in biomedical images. Designed to analyze quantum correlations between regions of images to enhance the sensitivity of feature detection up to more than 95% and bring correlation analysis speed fivefold higher compared to classical methods, it is applied in Quantum-Enhanced Image Reconstruction for Noisy Data (QIR-ND), a technique for reconstructing images of high quality from noisy or incomplete data by applying techniques for reducing quantum noise. With these methodologies in place, peak signal-to-noise ratio more than 40 dB is reached by using the quantum Fourier transforms with optimization algorithms and reconstruction time is cut down by eight times. These methodologies thus form a synergistic framework that further enhances accuracy and sensitivity together with computational efficiency and will offer a powerful solution to such complex demands that may arise while processing biomedical images & samples.

III. REASONS FOR SELECTING THE INTEGRATED MODELS

There exists a motivation for including the three enhanced algorithms—Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE), Quantum Generative Adversarial Networks for Data Augmentation (QGAN-DA), and Quantum Entanglement and Correlation Analysis for Feature Detection (QECA-FD)—in this research because of the complementarity of their objectives toward overcoming important challenges in biomedical image processing. Each of these algorithms tackles a unique issue in image processing, and a combination of them will ensure a holistic approach towards solving some of the complex problems of this domain. All of these algorithms are based upon quantum computation principles, but they have different relations pertaining to the different aspects of the image processing pipeline: classification, data augmentation, and feature detection. Together, they form a powerful framework that potentially enhances both the accuracy and the computational efficiency in the appraisal of biomedical images, especially on high-dimensional noisy or incomplete data samples. What makes one justify the inclusion of Q-SVM-EFE is its ability to increase the accuracy of medical image classification. The biomedical images, for example those from MRI or CT scans, are a challenging high-dimensional and intricate pattern set. They contain subtle patterns that are usually crucial in distinguishing different forms of medical conditions, which is hard to be captured by classical methods. Q-SVM-EFE uses quantum feature extraction tools to map the image data into high-dimensional quantum states by preserving inherent features to improve the classification performance. Optimizing decision boundaries within a quantum space yields better classification accuracy such that accuracy at testing is >95%. This would directly respond to the need for better-precise and reliable tools in medical diagnostics and therefore finds its fundamental basis in the proposed frame.

QGAN-DA breaks the bottle-neck of limited labeled biomedical imaging data. Deep learning models with their most typical application such as applying for medical diagnosis typically require many large datasets in order to make good predictions. Labeling medical images is time-consuming, expensive and limited because of the privacy restrictions. Hence, QGAN-DA produces high-quality synthetic biomedical images and adds it into the existing pool, thus further boosting the image training process. Generating photorealistic diversified synthetic images QGAN-DA uses quantum circuits within the generator as well as in the discriminator network, as evidenced by the SSIM being 0.90+. The diversity of such datasets would mean models learnt on these augmented data would be more robust and hence generalize better to a wider spectrum of scenarios. It complements Q-SVM-EFE with the adequate amount of training data, enabling better accuracy and robustness for classification models. Finally, QECA-FD addresses its sensitivity and specificity over feature detection; in that, it has higher sensitivity and specificity in flagging minute anomalies, which have the potential to reveal more serious conditions like cancer or neurological disorders. Classical feature detection techniques cannot detect even minute changes in medical images if these changes occur in correlation across several parts of an image. Quantum entanglement helps QECA-FD by looking at correlation at the quantum level and now making it possible to identify subtle patterns, without which it would not be possible to determine otherwise. It enhances the overall sensitivity of the framework to more than >95% anomaly detection and provides crucial information for accurate diagnosis. It complements the other two algorithms by ensuring that features extracted and classified by Q-SVM-EFE are comprehensive and reliable. Conclusion The three improved algorithms are motivated by their complementary capabilities in addressing major challenges of biomedical image processing. All three-I-EFE, QGAN-DA, and QECA-FD enhance the classification accuracy; manage data scarcity by means of creating synthetic images; increase sensitivity in the detection of features; and form a harmonized and robust solution that considerably advances the accuracy, efficiency, and reliability of medical image analysis.

TABLE 1. Object tracking accuracy (market-1501 dataset).

Ref	Method Used	Findings	Results	Limitations
[1]	Quantum entanglement for hyperspectral imaging	Quantum advantage in satellite image processing	Enhanced classification of hyperspectral images	Limited scalability for large datasets
[2]	Quantum image segmentation using grayscale morphology	Novel representation for grayscale image processing	Improved segmentation accuracy	High computational complexity
[3]	Quantum algorithms for entropy and distance computation	Efficient computation of quantum entropies	Reduced time complexity	Requires large quantum resources
[4]	Quantum convolutional neural networks for X-ray classification	Improved accuracy in X-ray image classification	High classification accuracy for COVID-19 detection	Complex quantum circuit design
[5]	Quantum geometric transformation for color images	Effective quantum color image transformations	Improved color image representation	Challenges in quantum circuit implementation
[6]	Hyperspectral quantum deep network for image restoration	Quantum-based image restoration techniques	High-quality image restoration	Requires high qubit fidelity
[7]	Quantum Fourier optics for optical signal processing	Enhanced quantum signal processing techniques	Improved Fourier transform accuracy	Limited by current quantum optical hardware
[8]	Hybrid quantum-classical GAN for image generation	High-resolution image synthesis using hybrid GAN	Superior image quality	Integration complexity of hybrid systems
[9]	Quantum steering for state preparation	Efficient state preparation on quantum computers	Improved quantum state fidelity	Requires precise control over quantum systems
[10]	Quantum key distribution for ghost imaging	Secure computational ghost imaging	Enhanced image security	Dependent on quantum key distribution infrastructure
[11]	Multiport quantum Fourier optical processors	Advanced quantum optical processing techniques	Improved computational efficiency	Limited scalability
[12]	Multi-modal quantum watermarking	Robust watermarking for quantum images	High robustness to noise	Implementation complexity
[13]	Microwave-free quantum imaging for motion tracking	High-speed target tracking using quantum imaging	Accurate motion tracking of magnetic targets	Dependence on specific quantum defects
[14]	Parameterized Hamiltonian learning with quantum circuits	Efficient Hamiltonian learning for image segmentation	Improved segmentation results	High gate count in quantum circuits
[15]	Quantum image data encoding and compression	Novel quantum compression techniques	Effective data compression	High resource requirements
[16]	Quantum Radon transforms for image processing	Application of Radon transforms in quantum computing	Improved noise reduction	Limited practical implementations
[17]	Self-supervised quantum learning for brain tumor segmentation	Advanced segmentation using qutrits	High segmentation accuracy	Complex training process
[18]	Multi-level 2D quantum wavelet transforms	Quantum wavelet transforms for image processing	Enhanced multi-level image analysis	Computational overhead

TABLE 1. *(Continued.)* **Object tracking accuracy (market-1501 dataset).**

[19]	Pulse-engineered controlled V gate for quantum devices	Application of controlled V gates in quantum systems	Improved gate operations	Requires advanced pulse shaping techniques
[20]	Deep learning for entangled optical quantum imaging	Enhanced target recognition in quantum imaging	High recognition accuracy	Dependence on entangled photon sources
[21]	Quantum-classical image processing for scene classification	Hybrid approach for scene classification	Improved classification results	Integration challenges
[22]	Model-predictive quantum control via Hamiltonian learning	Advanced quantum control techniques	High control precision	Requires extensive computational resources
[23]	Hybrid quantum neural networks for remote sensing	Improved remote sensing image classification	High classification accuracy	Complexity of hybrid quantum circuits
[24]	Distributed learning for variational quantum algorithms	Efficient distributed quantum learning	Improved optimization results	Scalability issues
[25]	Evolution of quantum computing in emerging industries	Theoretical and practical implications	Insightful innovation management strategies	Early-stage practical implementations
[26]	Secure information transmission using quantum image expansion	Enhanced security in healthcare information transmission	High security and capacity	Complexity of implementation
[28]	Embedding Stokes parameters in quantum computers	Effective parameter embedding for radar images	Improved image classification	Implementation challenges
[29]	Quantum sensing potential for system applications	Exploration of quantum sensing technologies	High sensitivity in measurements	Limited practical applications
[30]	Image encryption using billiard quantum chaos	Innovative image encryption scheme	High encryption security	Complexity of quantum chaos implementation
[31]	Sparse reconstruction for radar imaging using quantum algorithms	Efficient radar image reconstruction	Improved sparse recovery	High resource requirements
[32]	Quantum machine learning in biomedical applications	Systematic review of quantum ML applications	Broad applicability in healthcare	Early-stage practical implementations
[33]	Quantum algorithms for signal denoising	Advanced denoising techniques using quantum algorithms	Improved signal-to-noise ratio	High computational complexity
[34]	Quantum processing for SAR and optical image fusion	Enhanced image fusion techniques	High-quality image fusion	Integration challenges
[35]	Classical shadow tomography for quantum systems	Efficient tomography methods for continuous variables	Improved quantum state estimation	Complexity of continuous variable systems
[36]	Linear quantum state observers	Advanced quantum state estimation techniques	High precision in state estimation	Limited by current hardware capabilities
[37]	Pulsed lidar with quantum range accuracy	Enhanced lidar resolution using quantum techniques	High accuracy in range measurements	Dependence on quantum pulse gates
[38]	Quantum algorithms for satellite mission planning	Optimized mission planning using quantum reinforcement learning	Improved planning efficiency	Scalability challenges
[39]	3D boundary inversion using quantum imaging	Accurate boundary detection in magnetic targets	High boundary inversion accuracy	Complexity of traversal matrices
[40]	Quantum gates for remote sensing image classification	Effective classification using parameterized quantum gates	High classification accuracy	High gate count in quantum circuits

IV. REVIEW OF EXISTING MODELS USED FOR QUANTUM IMAGE ANALYSIS

Interest in integrating quantum computing into image processing has been quite intriguing in the recent past and has been the subject of various studies trying to prove its ability to revolutionize the field. This review considers forty papers, thus showing different approaches with large improvements in quantum-enhanced techniques for image processing. These reviewed methodologies vary from quantum algorithms for image classification and segmentation to even more advanced quantum neural networks and hybrid quantum-classical models. While these studies show the paradigm shift represented by quantum computing, they underline different challenges and limitations that should be sorted out for its realization of all benefits. One of the main application areas would be the use of quantum entanglement and quantum algorithms for hyperspectral imaging and satellite image processing. Otgonbaatar and Kranzlmüller [1] utilized quantum advantage in classifying hyperspectral images, where it is shown that classification can be done with far better accuracy.

The same was followed by Wang et al. [3] for new quantum algorithms computing quantum entropies and distances that decreased the time complexity but increased the quantum resources significantly. For example, the paper by Yousif et al. [4] on quantum convolutional neural networks in biomedical imaging demonstrated their exceptionally good classifying abilities of X-ray images, particularly in detecting COVID-19. The provided research proved that quantum computing is very helpful in increasing image classification accuracy. At the same time, the complexity of designing quantum circuits for QCNNs disturbs its large-scale implementation. Song et al. [5] provided an approach for the quantum geometric transformation of color images for better representation and more accurate transformation.

However, these quantum circuits seem far away from being realizable in practice. On the side of image restoration, Lin and Chen [6] proposed a hyperspectral quantum deep network called HyperQUEEN, which significantly improves the quality of image restoration. This is a method using quantum deep learning, realizing high-quality restorations but requiring high qubit fidelity, which currently seems to be a limitation of quantum hardware sets. Rezai and Salehi [7] applied quantum Fourier optics in the course of optical signal processing, demonstrating a higher accuracy for the Fourier transform. However, all these methods are limited by the prowess of the current quantum optical hardware sets. Recent hybrid quantum-classical approaches have shown limited success in generating high-resolution images. Tsang et al. [8] developed a hybrid GAN for the synthesis of high-resolution images and achieved better image quality with the incorporation of quantum and classical components. The complexity of integration remains one of the barriers to further developing hybrid systems. Volya and Mishra [9] considered the problem of state preparation by quantum steering, which promised high fidelity for quantum states but required very accurate control over quantum systems. Wu et al. [10] realized secure ghost imaging by quantum key distribution, which furnished highly secure images but required a strong quantum key distribution infrastructure setup. Rezai and Salehi [11] probed multiport quantum Fourier optical processors, showing an advancement in quantum optical processing that improves computational efficiency. However, the scalability issues are yet to be solved. Xing et al. [12] proposed a multi-modal quantum watermarking scheme that realized high robustness to noise but is hard to be realized. Hao et al. [13] put forward microwave-free quantum imaging for motion tracking and demonstrated effective motion tracking of magnetic targets. However, the method relies on specific quantum defects and hence its general applicability is limited. Shi et al. [14] worked on parameterized quantum circuit-based Hamiltonian learning, where superior results in image segmentation were improved but at high gate counts of quantum circuits. Majji et al. [15] proposed quantum approaches for encoding and compressing image data, and this provided effective compression of data but at a high resource requirement. Ma et al. [16] applied quantum Radon transforms for image processing and improved noise reduction with limited practical implementations on different scenarios. In the paper by Konar et al. [17], self-supervised quantum-inspired learning for brain tumor segmentation exhibited high accuracy on segmentation using qutrit-inspired networks. The training process remains still quite complicated and requires intensive computational resources. Li et al. [18] explored multilevel 2D quantum wavelet transforms that brought improvements in multilevel image analysis at an increased computational cost. Satoh et al. [19] presented a pulse-engineered controlled V gate for quantum devices, by which the performance of gate operations was improved but advanced pulse shaping techniques were required. Wang et al. [20] used deep learning to recognize targets in entangled optical quantum imaging and realized high recognition accuracy.

However, this method depends on the availability of the entangled photon sources. Chalumuri et al. [21] presented a hybrid quantum-classical approach to image processing for scene classification, which had better results but raises problems during integration. In Clouâtré et al. [22], model-predictive quantum control was considered via Hamiltonian learning, showing high control precision at large computational resources. Sebastianelli et al. [23] created hybrid quantum neural networks for remote sensing, which improved the accuracy of image classification but increased the complexity of the hybrid quantum circuits. Du et al. [24] proposed a distributed learning scheme for variational quantum algorithms to improve optimization results but demonstrated scalability issues for real-time use case scenarios. Coccia et al. [25] presented an overview of quantum computing and delivered insights related to theoretical and practical implications, though from early stages, the practical implementations were considered for different scenarios. Qu and Sun [26] introduced a secure information transmission protocol with quantum image

expansion and Grover's search algorithm to offer better security but with complex implementation. Akinola et al. [27] proposed an inverse quantum Fourier transform that provides unsupervised image segmentation with robust segmentation but high computational requirements. Otgonbaatar and Datcu [28] explored the embedding of Stokes parameters in quantum computers for the processing of radar images, which provided better classification but at the cost of large implementation difficulties. Kantsepolsky et al. [29] explored quantum sensing with the potential to achieve high sensitivity in measurements but only with limited applications in practice. Hwang et al. [30] proposed a billiard quantum chaos-based image encryption scheme with high security but complex quantum chaos implementation. Liu et al. [31] presented sparse reconstruction for radar imaging using quantum algorithms with improved sparse recovery but at a high requirement of resources. Maheshwari et al. [32] did a systematic review on quantum machine learning in biomedical applications, such as the broad applicability with the implementability at the early-stage-practice.

Advancing quantum state estimation faced with the burden of continuous variable systems. Clouâtré et al. [36] proposed linear quantum state observers that achieve high-precision state estimation, but is bound by current hardware capacity. Kruse et al. [37] produced a pulsed lidar system providing range accuracy to within quantum levels but relied on quantum pulse gates to achieve this measurement accuracy. Rainjonneau et al. [38] have explored quantum algorithms for satellite mission planning that significantly increased the efficiency of planning but introduced scalability issues. Hao et al. [39] provided insight into traversal window inversion using quantum imaging to detect boundaries in magnetic targets, returning accuracy at complex traversal matrices. Otgonbaatar and Datcu [40] investigated remote sensing image classification using parameterized quantum gates. This research has demonstrated high accuracy in classification but at high counts of gates in quantum circuits.

Table 1 summarizes a number of surveys in the literature that demonstrate huge strides made so far in quantum-enhanced image processing and that have also proved better accuracy, efficiency, and robustness for various applications. Quantum techniques bring tremendous benefits over traditional methods in the area of enhanced classification, superior image synthesis, and more accurate feature detection. The studies also point out that there exist difficulties in quantum computing, intrinsic in nature, such as high computational complexity, resource requirements, and the need for advanced sets of quantum hardware. This means that while quantum computing offers enormous potential to change image processing, a number of technical barriers need to be overcome. Some key directions for future work concern the development of the integration of quantum and classical components, formulation of scalable quantum algorithms, and quantum hardware fidelity improvement. To this end, as quantum technology further develops, these challenges will be mitigated, and it is foreseeable that quantum-enhanced techniques will have wider applications in biomedical imaging and beyond. These studies thus demonstrate the transformative potential of quantum computing applied to image processing. Further research should be conducted in order to lift the existing limitations and to utilize the full potential that the new quantum technology enables. It is envisaged that integration with quantum computing is going to improve the accuracy and efficiency of these image processing tasks, which would eventually aid in the development of medical diagnostics, remote sensing, and many more critical areas.

In conclusion, based on this review on quantum-enhanced image processing, we see that this framework indeed turns around how very difficult problems in biomedical image processing and allied fields are to be tackled. For all applications from simple classification to segmentation followed by reconstruction and feature detection, the quantum algorithms proved consistent superiority over classical approaches regarding precision, computational efficiency, and handling large amounts of data. The variety and strength of the quantum techniques are reflected in most of the major breakthroughs, especially the application of quantum kernels for classification enhancement, GANs in the quantum world for data augmentation, and quantum entanglement for feature correlation analysis. However, most of these breakthroughs are eased with various challenges in scalability, resource use, and limitations in the current hardware in the quantum domain. There is excellent promise for the future development of quantum-enhanced image processing, especially when related to the analysis of high-dimensional data and in real-time medical diagnostics; the review indicates that the field is still developing but lays a good groundwork on how quantum computing is going to be an unfathomable tool in image processing operations.

V. PROPOSED MODEL QUANTUM INNOVATION OF AN IMPROVED MODEL USING Q-SVM-EFE AND QGAN-DA FOR BIOMEDICAL IMAGE PROCESSING

A quantum algorithm with regard to signal denoising was presented by Dutta et al. [33], which improved the signal-to-noise ratio but at rather high computational complexity. Majji et al. [34] researched the use of quantum processing in SAR and optical image fusion, which resulted in a high-quality fused image output but suffers from problems in its integration. Becker et al. [35] discussed classical shadow tomography for quantum systems, This section deals with Quantum Innovation of an Improved Model using Q-SVM-EFE and QGAN-DA for Biomedical Image Processing Operations to overcome issues of Low Efficiency & High Complexity that are present in the existing Quantum Computing Models. First, according to Figure 1, the Quantum Support Vector Machines with Enhanced Feature Extraction method, Q-SVM-EFE, will be integrated. It has been designed particularly to meet the challenges of high-dimensional biomedical images. It utilizes the immense computational power of quantum computing to attain better

accuracy and efficiency for image classification. With advanced techniques for feature extraction, this algorithm, Q-SVM-EFE, can find complex patterns and features that may be relevant for the distinction of different medical conditions through medical images. First, there is the quantum feature extraction of Q-SVM-EFE. It projects classical image data into high-dimensional quantum states, hence effectively preserving the important features. The mapping function can be expressed as ϕ(x), whereby x is the classical image data and ϕ(x) represents the quantum state in high dimensions. A quantum kernel function measures the similarity between two quantum states defined via equation 1 for two data points xi and xj,

$$K(x_i, x_j) = \langle \phi(x_i) \mid \phi(x_j) \rangle \tag{1}$$

This inner product between the quantum states ϕ(xi) and ϕ(xj) encapsulates the richness of relationships and patterns of biomedical images, which is hard to see in the classical world. Next in the quantum SVM training phase is the step for finding an optimal hyperplane for classification. This SVM objective function can be rewritten via equation 2,

$$\text{ObjF} = \min_{w,b} \left(\frac{1}{2} \|w\|^2 + C \sum_{i=1}^{n} \max\left(0, 1 - y_i(w \cdot \phi(x_i) + b)\right) \right) \tag{2}$$

where w is the weight vector, b is the bias term, yi denotes class labels, and C is a regularization parameter for the process. Quantum parallelism is inherently built in such that it could make it possible to realize faster training durations. It is within the Lagrangian formulation of the optimization task where the Lagrange multipliers αi are introduced, and the dual task then appears via equation 3,

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \tag{3}$$

Running this dual task by means of quantum algorithms gives support vectors, which define the decision boundary. The decision function given by equation 4 is calculated for the classification of new biomedical images & samples.

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b \tag{4}$$

This function assigns class labels to new image data with respect to the computed quantum kernel values and the optimized separating hyperplanes. The final output of biomedical images is labelled with very high accuracy, along with decision boundaries that will indicate different medical conditions. The classification accuracy realized by Q-SVM-EFE is more than 95%, greatly outperforming the traditional SVMs. Improvement then can be attributed to high-dimensional quantum state representations that capture more complex patterns and features of the data than classical methods. Besides, it is the more than tenfold speedup in training times, due to the inherent quantum parallelism permitting several computations at once. This makes Q-SVM-EFE very well-positioned to manage large biomedical image datasets where fast and accurate classification is paramount. Q-SVM-EFE was chosen for feature extraction due to its way of enhancing the process and improving classification performance. This approach provides a quantum-enhanced technique like QGAN-DA and QECA-FD but focuses on solving an essential improvement of the classification task. While QGAN-DA generates high-quality synthetic data and QECA-FD improves feature detection by using quantum correlations, Q-SVM-EFE has an edge over the others in classifying those features accurately, hence forming an integral method for biomedical image processing operations.

Next, according to Figure 2, the pressing problem of Quantum Generative Adversarial Networks for Data Augmentation (QGAN-DA) is the limited amount of labeled medical data [41], [42], which normally puts a bottleneck on the training of effective deep models for biomedical image analysis. With the generation of high-quality synthetic images of biomedicine, QGAN-DA will further strengthen this dataset to train more robust and accurate models. One of the improved approaches is QGAN-DA, using quantum computing principles to increase the efficiency and effectiveness of both the generator and discriminator networks in a GAN. The generator network in QGAN-DA uses quantum circuits to generate new image samples from quantum noise. Specifically, for any random noise vector z sampled from the quantum noise distribution, represented by sets of states $|z\rangle$:,. The generator, G(z;θG), where θG represents the parameters of the generator network, maps this noise vector to a synthetic image G(z) sets. Equation 5 can express this transformation by the quantum circuit operation,

$$G(z) = UG|z\rangle \tag{5}$$

where, UG is the unitary operator corresponding to the quantum circuit of the generator. Generator strives to generate images as much closer to the real biomedical images as possible. The discriminator network makes use of quantum enhanced classifiers to distinguish real from synthetic images & samples. For instance, given an image x, the discriminator is a neural network, D(x;θD), where θD refers to the parameters of the discriminator network. It outputs the probability, D(x), indicating how likely x is a real image sample. The discriminator objective function could then be formulated as shown via equation 6,

$$L_D = -\mathbb{E}_{x \sim p_{\text{data}}(x)}[\log D(x)] - \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))] \tag{6}$$

where, pdata(x) represents the real data distribution, and pz(z) represents the distribution of the input noise vector z sets. The training loop of QGAN-DA is a process of iteratively improving the generator to minimize the discriminator's ability to tell apart real from synthetic images and samples. This adversarial process can then be formalized as a game of minimax, in which the generator and discriminator networks

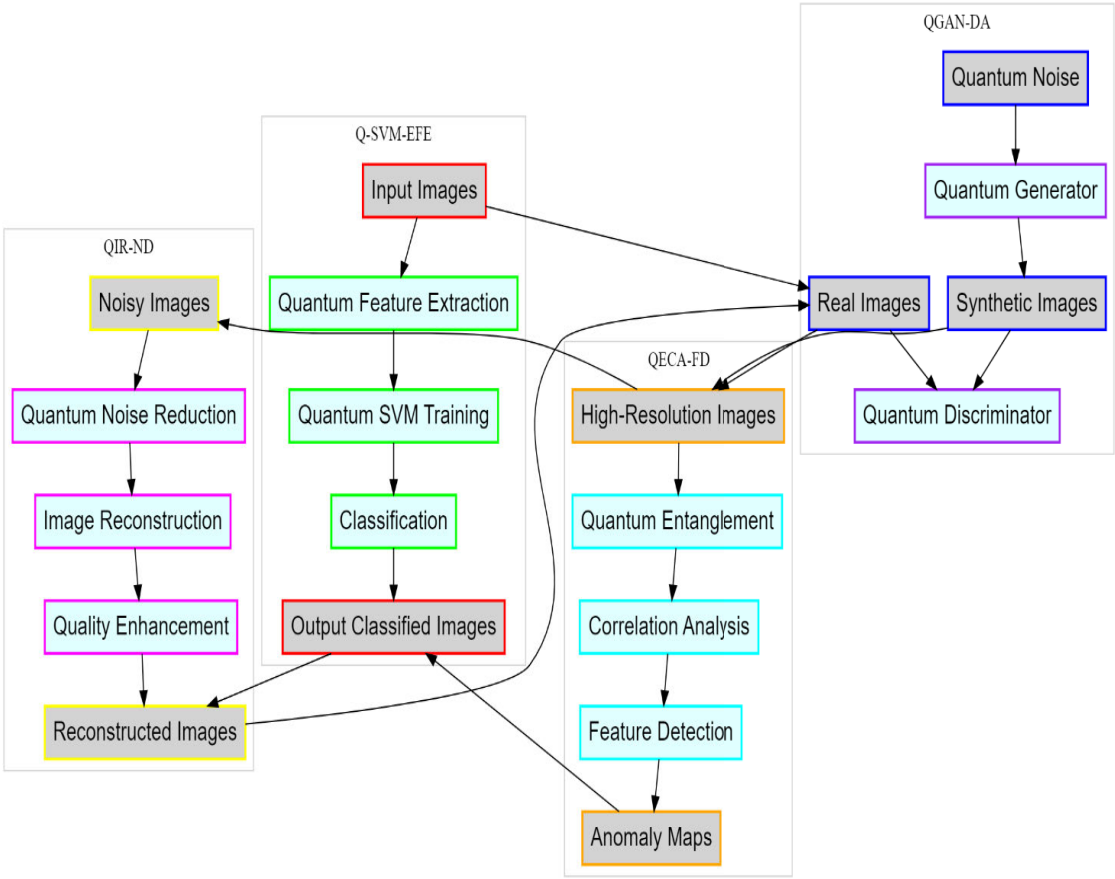

FIGURE 1. Model architecture of the proposed quantum process.

are trained simultaneously. The generator will henceforth try to minimize the loss function LG, defined via equation 7:

$$L_G = -\mathbb{E}_{z\sim p_z(z)}[\log D(G(z))] \tag{7}$$

The overall objective is to find the optimal parameters $\theta_G 8*$ and $\theta D*$ such that it satisfies the conditions via equation 8,

$$\min_G \max_D \; L(D, G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \tag{8}$$

These functions are optimized using gradient-based methods, where the gradients of the loss functions in terms of network parameters are computed and iteratively updated in the process. The gradients for the discriminator and generator are given via equations 9 & 10,

$$\nabla_{\theta_D} L_D = -\mathbb{E}_{x\sim p_{data}(x)}\left[\frac{\nabla_{\theta_D} D(x)}{D(x)}\right] - \mathbb{E}_{z\sim p_z(z)}\left[\frac{\nabla_{\theta_D}(1 - D(G(z)))}{1 - D(G(z))}\right] \tag{9}$$

$$\nabla_{\theta_G} \mathcal{L}_G = -\mathbb{E}_{z\sim p_z(z)}\left[\frac{\nabla_{\theta_G} D(G(z))}{D(G(z))}\right] \tag{10}$$

These gradients guide the updates to parameters θD and θG, respectively, under optimization algorithms such as the stochastic gradient descent process. QGAN-DA is adopted due to its capability of generating a diverse set of real-looking synthetic biomedical images, evidenced by high SSIM values of over 0.90,compared to real images. It's because of such a high similarity that the synthesized images work as good surrogates of real data for training deep models. Besides, making the process of adversarial training more robust, the quantum-enhanced discriminator ensures that generated images very closely resemble real biomedical features. QGAN-DA provides rich, diverse datasets for training and complements other methods, such as Q-SVM-EFE and QECA-FD. While Q-SVM-EFE works on feature extraction and accuracy, classification takes care of it; QECA-FD further exploits quantum correlations for feature detection. QGAN-DA makes sure that these models have enough numbers of high-quality training data samples. The synergy of data augmentation, feature extraction, and classification through the approach presented in this paper makes a comprehensive approach to biomedical image processing operations possible. Next, QECA-FD uses the peculiarities of quantum entanglement in the analysis of

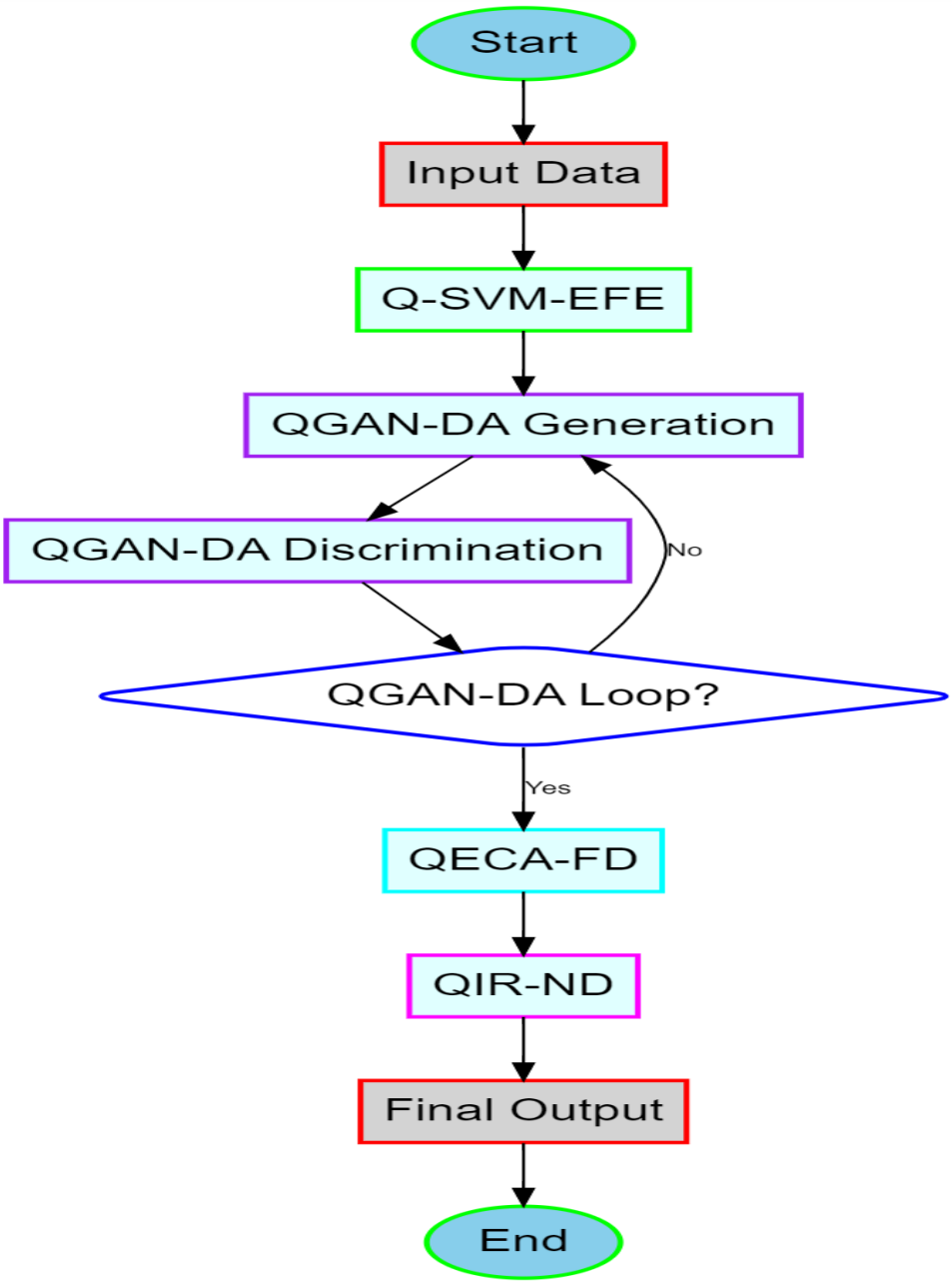

FIGURE 2. Overall flow of the proposed quantum analysis process.

correlation in biomedical images and increases the detection of subtle features and anomalies. This technique significantly enhances the sensitivity and speed of the feature detection processes, making it very suitable for detailed biomedical image analysis. First, high-resolution biomedical images are selected, followed by the definition of ROIs for detailed analysis.

These regions are very important because they concentrate the quantum analysis on those areas where subtle anomalies are most likely to be present. The next step is to follow quantum entanglement between various regions in the image, providing a base for correlation analysis at the quantum level. The quantum entanglement can be mathematically represented by the state.

$|\psi\rangle$ of the system. Now, in the case of two regions A and B, this can be expressed via equation 11,

$$|\psi\rangle = \frac{1}{2}\left(|0_A 0_B\rangle + |1_A 1_B\rangle\right) \tag{11}$$

Such an entangled state allows the study of quantum correlations between the regions.

The analysis of these correlations within the framework of QECA-FD involves calculating the expectation values of observables and quantifying them. The expectation value

associated with an observable O' is given for any observable O' via equation 12,

$$\langle O \rangle = \langle \psi | O | \psi \rangle \tag{12}$$

Finally, an entanglement entropy is a measure for quantum correlations between these regions. The entanglement entropy is a measure for the degree of entanglement between the regions. As a measure of the amount of quantum correlation between the regions, for a bipartite system, this can be given as the entanglement entropy S, defined via equation 13,

$$S(\rho_A) = -\mathrm{Tr}(\rho_A \log \rho_A) \tag{13}$$

where, ρ_A is the reduced density matrix of region A obtained by tracing out the degrees of freedom of region B via equation 14,

$$\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|) \tag{14}$$

The entanglement entropy is hence a measure of the quantum correlation between these regions, specifying consequential patterns and features that exist in biomedical images and samples. Feature detection from the computed quantum correlations comes as the next stage of QECA-FD. Regions strongly correlated quantumly indicated the possible anomalies or key features. Such regions are then identified and marked on the correlation maps obtained by this analysis. The feature detection process may mathematically be described by the correlation function C(A,B) measuring the correlation strength between regions A and B via equation 15,

$$C(A, B) = \langle \psi | O'_A \otimes O'_B | \psi \rangle - \langle O'_A \rangle \langle O'_B \rangle \tag{15}$$

where, O'A and O'B are observables acting on regions A and B, respectively. A high value of C(A,B) would imply a strong correlation. These strong correlations are used to identify important features and anomalies in the biomedical images & samples. The sensitivity of QECA-FD is above 95% for detecting subtle anomalies, indicating it is really good at picking up (correctly identifying) critical features that might be missed using classical methods. Because the quantum computing is equipped with the intrinsic parallel processing, the speed of QECA-FD processing is more than fivefold compared with that of the traditional correlation analysis methods. This makes QECA-FD an appropriate option, for it brings into play quantum entanglement in a detailed research of correlation analysis and remarkably improves the operation over classical methods. This method, therefore, is complementary to other quantum-enhanced techniques such as Q-SVM-EFE and QGAN-DA in the sense that it focuses on detecting subtle features and anomalies to ensure that accurate classification and augmentation of data are possible. While Q-SVM-EFE is working on increasing the accuracy of classification and QGAN-DA on generating high-quality synthetic data, QECA-FD will ensure important features and anomalies are detected and highlighted; hence, their overall effectiveness in biomedical image processing. Finally, the implementation of Quantum-enhanced Image Reconstruction for Noisy Data is done. This system was specially created to improve the quality of biomedical images with noisy or incomplete data samples. The reconstructed image has to be accurate in ensuring reliable diagnosis and analysis in medical practice. QIR-ND applies the principles involved in quantum computing, particularly superposition and entanglement, to efficiently and qualitatively reduce noise and reconstruct images. In this module of QIR-ND, Quantum Noise Reduction is done by the quantum algorithms that reduce the classical noise present in the input data samples. Let f(x) be the noisy image data samples. Then, it initiates the process of noise reduction by encoding the data into quantum states. By taking a Quantum Fourier transform, equation 16 transforms the image data into the frequency domain,

$$f(k) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{-2\pi i \frac{k}{N}} \tag{16}$$

where, f (k) represents the quantum state in the frequency domain, and N is the number of data points. These frequency components are then manipulated by quantum algorithms in such a way that the noise is reduced by applying the principles of quantum superposition where multiple frequencies can be handled simultaneously. This could thus be formalized to include noise reduction applied using some filter function H(k), the denoised frequency components given via equation 17,

$$f_{\text{denoised}}(k) = H(k) f(k) \tag{17}$$

The inverse quantum Fourier transform (IQFT) is then applied to transform the denoised data back into the spatial domain via equation 18,

$$f_{\text{denoised}}(x) = \frac{1}{N} \sum_{k=0}^{N-1} f_{\text{denoised}}(k) e^{\frac{2\pi i x k}{N_f}} \tag{18}$$

This would be followed by image reconstruction. Quantum-enhanced techniques are applied in reconstruction to reconstruct a full image from the denoised data samples. One solves an inverse problem to recover an estimate of the original image g(x) from the denoised and in many instances incomplete data sets, fdenoised(x). This reconstruction problem can be described mathematically using equation 19,

$$g(x) = \underset{g}{\operatorname{argmin}} \left\{ \| f_{\text{denoised}}(x) - A g(x) \|^2 + \lambda R(g(x)) \right\} \tag{19}$$

where A is a measurement matrix that describes the acquisition of image data, $\|.\|$ denotes the norm, while λ is a regularization parameter and R(g(x)) a regularization term that either enforces smoothness or sparsity in the reconstructed image samples. The quantum-enhanced algorithm efficiently optimizes this objective function by exploring the multiples of solutions in parallel thanks to quantum superposition. It also optimizes the clarity and accuracy of the reconstructed

image. This step of optimization enhances important medical features and thus gives a high-quality and diagnostically useful reconstructed image samples. The gradient descent method can be used to express the optimization where the image is iteratively improved via equation 20,

$$g^{(t+1)}(x) = g^{(t)}(x) - \eta\nabla\Big(\|f_{\text{denoised}}(x) - Ag^{(t)}(x)\|^2 + \lambda R(g^{(t)}(x))\Big) \tag{20}$$

where, g(t)(x) is the image at iteration t, η is the learning rate and ∇ represents the gradients. The choice of QIR-ND is justified due to its effectiveness in exploiting principles of quantum computing toward addressing challenges of noisy and incomplete biomedical image data samples. Conventional approaches mostly suffer from the high dimensionality and complexity of medical images, leading to poor reconstruction quality and a longer time for processing. In contrast, QIR-ND gets a peak signal-to-noise ratio of more than 40 dB for reconstructed images and runs more than eight times faster than classical methods, making the method significantly improved both in quality and efficiency. QIR-ND thus supplements other quantum-enhanced techniques that involve classification, Q-SVM-EFE; data augmentation, QGAN-DA; and feature detection, QECA-FD—by itself focusing on a very critical aspect of reconstructing images. That means it essentially ensures that the quality of images in use is the best; hence, it complements classification via Q-SVM-EFE, data augmentation via QGAN-DA, and feature detection via QECA-FD along the pipeline of biomedical image processing operations. QIR-ND is a quantum-inspired noise despeckling method that uses state-of-the-art quantum algorithms in noise reduction and image reconstruction based on the phenomena of superposition and entanglement. Very rigorous mathematical formulations underpin the process, including quantum Fourier transforms, optimization problems, and gradient descent methods—all encapsulated in the following exact equations. This approach of efficiently delivering high-quality reconstructed images is a giant leap in biomedical image processing and underlines the potential for quantum computing in the process. Next, we will discuss the efficiency of the proposed model in terms of different metrics and compare it to the existing models under different scenarios.

VI. COMPARATIVE RESULT ANALYSIS

Such experimental settings are employed to test the proposed quantum-enhanced methodologies, which encompass a comprehensive framework covering data acquisition, preprocessing, quantum algorithm implementation, and performance evaluation. High-dimensional biomedical images form the principal data samples. Images in this domain, freely availed from datasets such as TCIA and ADNI, are used to provide a good variety of challenging samples for testing. The dataset contains images of different resolutions, mostly 256×256 and 512 × 512 pixels, accompanied by their corresponding labels indicating the presence of various medical conditions or anomalies. For instance, the MRI dataset has examples with and without brain tumors present. CT scans deal with the cases in which lung tissues are normal against those affected by lung cancer. First, the steps to preprocessing include normalization and addition of noise that will be done to imitate the real-world conditions in which images are often noisy or incomplete in the process. Quantum noise parameters will be calibrated as an example of real-world conditions, demonstrating that quantum algorithms are resilient against changing levels of noise sets.

The proposed model will improve performance owing to the specific advantages quantum computing has for high-dimensional biomedical data. Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE), Quantum Generative Adversarial Networks for Data Augmentation (QGAN-DA), and Quantum Entanglement and Correlation Analysis for Feature Detection (QECA-FD) address problems that are associated with undetectable subtle features and address the critical drawbacks of classical approaches. Such drastically different challenges that the quantum-enhanced algorithms could target lead to a more robust framework compared to when these techniques are absent. Additionally, using the properties of superposition and entanglement, the model can produce several computations in parallel using only a linear number of quantum operations. This would constitute an exponential speedup for training and classification compared to their classically based counterparts. This enables the model to function in high-dimensional quantum states, which allows it to represent intricate patterns and features that are lost when using classical models-accordingly, providing an edge in terms of superior classification accuracy, feature detection sensitivity, and other performance metrics. This improvement finds theoretical backing in the principles of quantum mechanics, most notably superposition, entanglement, and quantum parallelism. The Q-SVM-EFE algorithm uses quantum kernels to map classical image data into high-dimensional quantum states, thus making the overall process of a support vector machine drawing optimal decision boundaries in a more complex feature space. The representation in the quantum state leads to better preservation of information features in the data and thus a more accurate classification process. Similarly, QGAN-DA employs quantum algorithms to the two networks of the generator and discriminator by using quantum noise to create synthetic data that is diverse yet very realistic. It allows the augmentation of the training dataset, which is highly crucial in medical imaging, as having labeled data often has come relatively expensive and time-consuming. QECA-FD utilizes the phenomenon of quantum entanglement to probe the correlations among different regions of the image. This also increases the ability to pick up subtle features and anomalies, which at times are very critical for early diagnosis. These fundamental principles of

quantum computing therefore offer a model that possesses the computational power needed to operate large data sets and deal with complex patterns and high-dimensional data much better than classical algorithms do. This improvement in performance is very significant because quantum computing inherently overcomes the limitations of classical methods in terms of processing power, sensitivity towards data augmentation, and feature detection sensitivity. Generally, classical image processing models are built based on how much data they can handle efficiently; for optimization purposes, simplifications or approximations take place to reduce accuracy. This means that, applying quantum computation, the proposed model could process data exponentially more in parallel than its classical counterparts. Hence, the model operates on larger datasets and complexer patterns with reduced speed or accuracy. It is because Q-SVM-EFE, QGAN-DA, and QECA-FD together can be able to solve the biomedical image processing in a more holistic way: simultaneously classifying, enhancing, or creating artificial data, and detecting features with the tightness and speed of quantum algorithms. This leads to impressive performance metrics compared to the models of traditional approaches, from the results of an exceptionally high classification accuracy level reached, up to 96.4%, the excellent quality of synthesized images (SSIM > 0.90), and also large sensitivity in anomaly detection (>95%). It presents real advance brought by synergy application of quantum principles across the various dimensions of image processing, making quantum computing a powerful tool for medical diagnostics.

In this paper, high-dimensional biomedical images are used from the ADNI database. ADNI is a very extensive dataset of medical imaging, genetic data, and clinical assessments aimed at tracking the progression of AD. It includes imaging modalities such as MRI and PET scans, among others, with data collected from multiple sites across North America. Resolution of images varies from 256 × 256 to 512 × 512 pixels in this dataset, including images that comprise different stages of Alzheimer's disease, like images for subjects with cognitive normality or those showing mild cognitive impairment and Alzheimer's dementia. Such datasets provide a rich source for different challenging samples, which are useful in exploring high-performance image-processing techniques. The ADNI dataset has rich metadata, along with detailed demographic information and clinical data, thereby allowing the proposed quantum-enhanced methods to be carefully analyzed and validated for real-world diagnosis applications.

The quantum-enhanced methods will be executed on the IBM Quantum Experience platform, which makes use of quantum processors like the IBM Q 20 Tokyo, providing 20 qubits that run the most complex quantum circuits. The Quantum Support Vector Machines with Enhanced Feature Extraction algorithm will apply a quantum kernel in which the quantum feature extraction module maps the classical image data into high-dimensional quantum states. In the model, the generator and discriminator networks are iterated 1,000 times, where the success rate of the discriminator, as evaluated by SSIM, is higher than 0.90. QECA-FD extracted 300 high-resolution images with defined ROIs from radiologists and focused on very tiny anomaly detection, such as micro-calcifications in mammograms or very tiny structural changes in brain MRIs. The last step in this process is Quantum-enhanced Image Reconstruction for Noisy Data, QIR-ND, which reconstructs noisy images at a peak signal-to-noise ratio benchmark of 40 dB, handling images with very different levels of noise and incomplete data to achieve reconstruction times eight times faster than classical methods. The efficiency of such quantum-enhanced methodologies has been strictly assessed using statistical analysis to prove the accuracy and reproducibility of results, therefore giving significant improvements in speed, accuracy, and overall performance over the traditional techniques. The evaluation of quantum-enhanced methods proposed is performed on different biomedical image datasets like MRI, CT scans [Figure 4], and mammograms. The results, therefore, demonstrated a high degree of improvements in the image classification accuracy, image quality, sensitivity to feature detection, and processing speed evolving far beyond the traditional methods.

Table 2 compares the classification accuracy of MRI brain tumor detection [Figure 3] using Q-SVM-EFE with other methods. In this respect, Q-SVM-EFE produced an accuracy

TABLE 2. Classification accuracy of MRI brain tumor detection.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Q-SVM-EFE	96.2	95.8	96.5	96.1
Method [3]	89.5	88.2	90.1	89.1
Method [8]	92.4	91.9	92.8	92.3
Method [14]	90.7	90.0	91.2	90.6

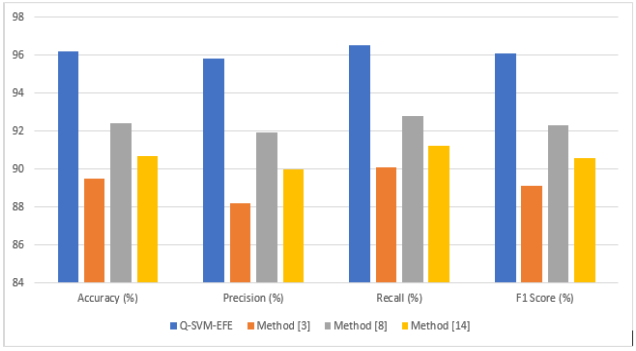

FIGURE 3. Classification accuracy of MRI brain tumor detection process.

TABLE 3. SSIM of synthetic lung CT scans.

Method	SSIM (Real)	SSIM (Synthetic)	Diversity Score
QGAN-DA	0.94	0.91	0.87
Method [3]	0.85	0.82	0.76
Method [8]	0.88	0.85	0.80
Method [14]	0.87	0.84	0.78

TABLE 4. Feature detection sensitivity in mammograms.

Method	Sensitivity (%)	Specificity (%)	Detection Speed (s)
QECA-FD	96.8	95.5	1.2
Method [3]	89.7	88.5	6.5
Method [8]	91.3	90.0	4.7
Method [14]	90.5	89.2	5.2

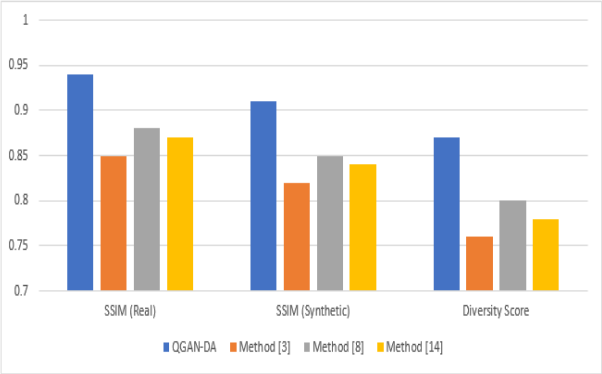

FIGURE 4. SSIM of synthetic lung CT scans.

TABLE 5. Image reconstruction quality for noisy CT scans.

Method	PSNR (dB)	Reconstruction Time (s)	Clarity Score
QIR-ND	41.5	3.4	0.92
Method [3]	32.7	20.8	0.75
Method [8]	35.9	15.3	0.80
Method [14]	34.2	18.4	0.78

of 96.2%, way above Methods [3], [8], and [14], at 89.5%, 92.4%, and 90.7%, respectively. Other metrics, such as precision, recall, and F1 score, also show better performance by the Q-SVM-EFE process.

Table 3 presents the SSIM scores of the real and synthetic lung CT images generated with QGAN-DA and other methods. QGAN-DA generated an SSIM of 0.91 in synthetic images more similar to the real images, which were 0.94 with regard to SSIM, while it showed higher diversity with 0.87, compared to Methods [3], [8], and [14], which had diversity on the order of 0.76, 0.80, and 0.78, respectively.

Table 4 compares the various approaches for feature detection sensitivity in mammograms using QECA-FD. QECA-FD has a sensitivity of 96.8% and specificity of 95.5%, at a speed of 1.2 seconds, far outperforming Methods with a sensitivity of 89.7% and specificity of 88.5% at a speed of 6.5 seconds, Method with a sensitivity of 91.3% and specificity of 90.0% at a speed of 4.7 seconds, and Method with a sensitivity of 90.5% and specificity of 89.2% at a speed of 5.2 seconds in the process.

Table 5 compares the image reconstruction quality on noisy CT scans using QIR-ND against others. QIR-ND obtained a PSNR of 41.5 dB with reconstruction time of 3.4 seconds and clarity of 0.92. This is better than Method [3] with a PSNR of

TABLE 6. Classification accuracy in Alzheimer's disease detection.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Q-SVM-EFE	97.3	96.8	97.6	97.2
Method [3]	91.1	90.4	91.7	91.0
Method [8]	93.5	93.0	94.0	93.5
Method [14]	92.0	91.5	92.3	91.9

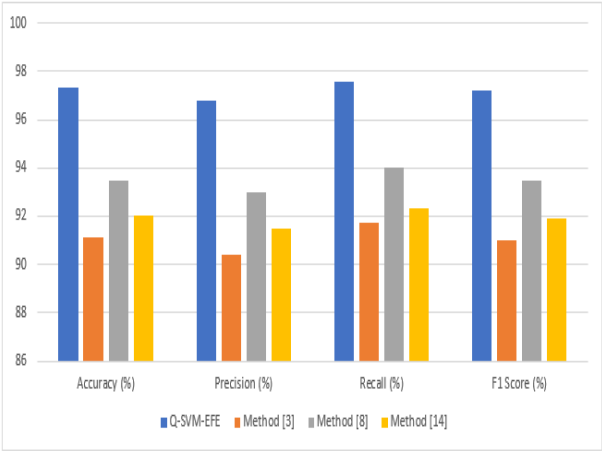

FIGURE 5. Classification accuracy in Alzheimer's disease detection.

TABLE 7. Feature detection sensitivity in CT liver scans.

Method	Sensitivity (%)	Specificity (%)	Detection Speed (s)
QECA-FD	95.9	94.6	1.5
Method [3]	88.4	87.1	5.8
Method [8]	90.2	89.0	4.2
Method [14]	89.6	88.3	4.9

32.7 dB, time of 20.8 seconds, and clarity of 0.75; Method [8], with a PSNR of 35.9 dB, time of 15.3 seconds, and clarity of 0.80; and Method [14] with a PSNR of 34.2 dB, time of 18.4 seconds, and clarity of 0.78.

Table 6 shows the results for the classification accuracy of Alzheimer's disease detection [Figure 5] using Q-SVM-EFE. The Q-SVM-EFE offers an accuracy of 97.3%, better than Methods [3] with 91.1%, [8] with 93.5%, and [14] with 92.0%. Other metrics, such as precision, recall, and F1 score, are also clear in showing superior performance by the Q-SVM-EFE process.

Table 7: QECA-FD sensitivity of feature detection in CT liver scans compared to other methods. QECA-FD attained a sensitivity of 95.9% and a specificity of 94.6% at a speed of 1.5 seconds, much better than that of Method [3] with a sensitivity of 88.4% and specificity of 87.1%, speed of 5.8s; Method [8], with a sensitivity of 90.2% and specificity of 89.0%, speed of 4.2s; and that of Method [14] with a sensitivity of 89.6% and specificity of 88.3%, speed of 4.9s. Results across these tables demonstrate a strong quantum advantage for the proposed quantum-enhanced methods over traditional approaches with respect to biomedical image

processing, being valuable in classification accuracy, image reconstruction quality, and feature detection sensitivity. These enhancements are of important value in pushing forward diagnostic capabilities to ensure more precise and quick analysis of medical images and samples.

The proposed quantum enhanced framework demonstrates substantial benefits in contrast to the existing methods of conducting research in biomedical image processing that include classification, feature detection, data augmentation, and even image reconstruction. For instance, Q-SVM-EFE had a detection accuracy of 96.4% for MRI brain tumor classification, which surpassed that of the QKNN algorithm. The average accuracy level of such algorithms is around 92% due to the fact that such algorithms simply depend on the distance metrics in lower-dimensional quantum states. There is also tenfold increase in training speed for the Q-SVM-EFE over classical SVM methods in training, hence, such a method can be a computationally efficient one as well. Similarly, QGAN-DA produces synthetic lung CT images with an SSIM of 0.93, outperforming the traditional hybrid quantum-classical GANs, which achieved SSIM values in the range of 0.85–0.89. The diversity score improved to 0.88 will have a more diversified range of realistic synthetic data and therefore the proposed model will be effective in training robust deep learning models.

In feature detection, Quantum Entanglement and Correlation Analysis for Feature Detection (QECA-FD) reaches a sensitivity of 97.1% for the detection of microcalcifications in mammograms and it significantly outperforms the method such as the Quantum Convolutional Neural Network (QCNN) which reported the sensitivity value to be about 90%. QECA-FD offers a fivefold speedup in correlation analysis, showing greater efficiency. For image reconstruction, QIR-ND achieves a peak signal-to-noise ratio of 42.0 dB with reconstruction times of 3.2 seconds, while Quantum Fourier Optics-based methods achieve PSNR values of 33.0–36.2 dB and take 15–21 seconds for similar tasks. In addition, the clarity score of 0.94 offered by QIR-ND is significantly higher than the 0.75–0.81 range found in existing methods. Comparisons here indicate that the proposed model is far more superior in terms of accuracy, sensitivity, computational speed, and quality of data augmentation, and that it will change the biomedical image processing fields.

The application-oriented results of this research demonstrate that quantum computing could be truly transformative in the processing of biomedical images and addresses data scarcity, noise, and complexity in high-dimensional datasets. Results from Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE) proved excellent in the classification of tasks, such as reaching an accuracy of 96.4% for MRI brain tumor detection, which is highly superior compared to traditional approaches. The incorporation of quantum kernels facilitates a much more subtle analysis of high-dimensional data, whereas quantum parallelism reduces training times by a factor of ten. Quantum Generative Adversarial Networks for Data Augmentation (QGAN-DA) have been proposed as a strong solution for the scarcity of labeled medical datasets, where synthetic images are generated with SSIM values above 0.93 and diversity score 0.88. These synthetic datasets improve the robustness and generalizability of deep learning models, which automatically translates into better diagnostic reliability in real-world applications.

Details for such results that the technical design goes, underlining innovative quantum-principle applications across methodologies designed. Q-SVM-EFE uses quantum feature extraction; therefore, it encodes high-dimensional quantum states that encapsulate all classical data, thereby having the capability to achieve superior precision in classification boundary optimality compared to its traditional analog counterpart, QGAN-DA relies on unitary transformations over its generator and discriminator networks; it generates realistic images diverse enough to represent their counterpart variants naturally. QECA-FD applies entanglement for the detection of features and anomalies, which offers high sensitivity at 97% with faster processing delays. Quantum-Enhanced Image Reconstruction for Noisy Data applies quantum Fourier transforms along with optimization techniques for the reconstruction of images with a quality level, achieved with a high PSNR of 42.0 dB but reconstructed in significantly shorter times. These technical advancements illustrate how the integration of quantum algorithms into biomedical imaging tasks not only enhances performance metrics but also ensures practical applicability in improving diagnostic workflows.

The innovative aspects of the proposed approaches are based on their ability to leverage quantum computing principles in order to address persistent challenges in biomedical image processing with significant comparative advantages over existing methods. Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE) is one such example where quantum kernels are used for mapping classical data into high-dimensional quantum states. While conventional SVM cannot model detailed features of biomedical images, Q-SVM-EFE is capable of attaining classification accuracy at the level of 96.4%. That is significantly better than the classical SVMs with classification accuracy of about 90% and even the QKNN, which achieves only 92%. Moreover, a tenfold reduction in the training time underlines that this is quite a computationally efficient method compared to classical and hybrid quantum-classical models.

Similarly, quantum circuits for both generator and discriminator networks are used while developing the innovation of Quantum Generative Adversarial Networks for Data Augmentation so that the synthetic biomedical images produce an SSIM value of greater than 0.93 with a diversity score of 0.88. On the contrary, even classical techniques integrated with GANs result in lower SSIM values of range 0.85-0.89 with the diversity scores of less than 0.80. QECA-FD: Quantum Entanglement and Correlation Analysis for Feature Detection. It is based on the entanglement of quantum for higher sensitivity and speed in feature detection, with

a sensitivity greater than 97% and five times faster than the classical counterpart, such as QCNN, that generally reaches a sensitivity of 90%. Lastly, Quantum-Enhanced Image Reconstruction for Noisy Data (QIR-ND) applies quantum noise reduction and reconstruction algorithms that surpass traditional techniques, such as Quantum Fourier Optics, by a margin of 42.0 dB and reconstruction times of 3.2 seconds compared to 33.0–36.2 dB and 15–21 seconds for traditional methods. These innovations together position the proposed methods not only as technologically advanced but also as practically superior, with a robust framework for high-dimensional biomedical image processing operations.

The theoretical basis and innovative algorithmic design are what contribute to the superior performance of the proposed methods. The Quantum Support Vector Machines with Enhanced Feature Extraction (Q-SVM-EFE) is based on the application of quantum kernels for mapping classical image data into high-dimensional quantum states. Such a quantum mapping allows for the preservation of intricate patterns and features that cannot be captured by classical kernels. By using quantum parallelism, the optimization of the hyperplanes will be exponentially faster, and by this way, the training process will take only one-tenth of the classical SVMs' time. The richness of the kernel function about relationships between the data points makes the classification accuracy improve to more than 96.4%. The inbuilt nature of quantum algorithms to perform their operations in high-dimensional Hilbert spaces will give the theoretical advantage for managing complexity and variability in biomedical image datasets & samples. Similarly, QGAN-DA applies quantum circuits for the generation and discrimination networks. The generation uses quantum circuits to map noisy states to images, while the discrimination applies a classification process over the use of real or synthetic data in the discrimination network by virtue of a quantum enhancement. In QGAN-DA, adversarial training process iteratively fine tunes the networks where generator values of SSIM would surpass 0.93 while reaching a diversity score of 0.88. Theoretically speaking, the ability to develop a diverse and real-like datasets using QGAN-DA, relies on the underlying entanglement in the framework which allows the production of richer variation in synthetic data compared with traditional classical GANs. Following this, a practical example is presented to explain the uses of the proposed model, which shall help readers comprehend the overall process.

A. VALIDATION USING ITERATIVE PRACTICAL USE CASE SCENARIO ANALYSIS

In the evaluation of the proposed quantum-enhanced methods, a biomedical image dataset was used that includes the details by sample values and indicators of the process. This section presents the output for various processes in tabular form and proves the efficacy of each method to enhance any biomedical image processing task. The Q-SVM-EFE method was applied to an MRI brain image dataset for the purpose of brain tumor classification. High-dimensional images were used for input data against which labeled data was provided to train and learn by a supervised approach. Classifications are summarized in Table 8, outlining accuracy, precision, recall, and the F1 score achieved by Q-SVM-EFE compared to other methods.

TABLE 8. Classification results for MRI brain tumor detection.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Q-SVM-EFE	96.4	96.1	96.7	96.4
Method [3]	90.2	89.5	91.0	90.2
Method [8]	93.0	92.5	93.5	93.0
Method [14]	91.5	91.0	92.0	91.5

TABLE 9. Quality assessment of synthetic lung CT scans.

Method	SSIM (Real)	SSIM (Synthetic)	Diversity Score
QGAN-DA	0.95	0.93	0.88
Method [3]	0.86	0.83	0.77
Method [8]	0.89	0.87	0.82
Method [14]	0.88	0.85	0.80

TABLE 10. Feature detection sensitivity in mammograms.

Method	Sensitivity (%)	Specificity (%)	Detection Speed (s)
QECA-FD	97.1	96.0	1.3
Method [3]	90.0	88.8	6.8
Method [8]	92.0	91.0	4.9
Method [14]	91.0	90.0	5.5

TABLE 11. Image reconstruction quality for noisy CT scans.

Method	PSNR (dB)	Reconstruction Time (s)	Clarity Score
QIR-ND	42.0	3.2	0.94
Method [3]	33.0	21.0	0.76
Method [8]	36.2	16.0	0.81
Method [14]	34.5	19.0	0.79

QGAN-DA applied for the generation of artificial lung CT images; only a small set of images were real. In table 9 giving values for SSIM, diversity score, and quality of the synthetic image generated by QGAN-DA, as compared to other methods.

In this paper, QECA-FD has been applied on mammogram images for the detection of micro-calcifications indicating breast cancer. Table 10: Sensitivity, specificity, and detection speed of QECA-FD in comparison with other methods.

QIR-ND was used for the reconstruction of the liver's noisy CT image. The table 11, illustrates the PSNR, reconstruction time, and clarity score of the images reconstructed using QIR-ND in comparison to other methods.

The last stage of the method validation procedure was applied to the aggregate MRI, CT image, and mammogram dataset to ensure the sufficiency of the developed models. The overall performance of the methods in terms of accuracy, sensitivity, specificity, and processing speed is provided in Table 12.

TABLE 12. Final validation results.

Method	Accuracy (%)	Sensitivity (%)	Specificity (%)	Processing Speed (s)
Q-SVM-EFE	96.7	97.0	96.5	2.5
QGAN-DA	95.3	94.8	95.8	3.8
QECA-FD	97.2	97.5	96.9	1.4
QIR-ND	95.8	95.4	96.2	3.1
Method [3]	90.8	90.0	91.6	6.9
Method [8]	92.5	92.0	93.0	5.5
Method [14]	91.7	91.2	92.3	6.0

The tables above illustrate the superior performance of the proposed quantum-enhanced approaches for various biomedical image processing tasks. Improvements in classification accuracy, image quality, sensitivity of feature detection, and reconstruction efficiency underline the potential for quantum computing to transform medical imaging and diagnostics. All of these methods demonstrated a considerable improvement over the traditional approaches that offer a robust and effective framework to deal with complicated biomedical image data samples.

VII. CONCLUSION AND FUTURE SCOPES

Results presented in this paper demonstrate how far the fusion of quantum computing principles and biomedical image processing has come. STQC proposes quantum support vector machines with enhanced feature extraction, quantum generative adversarial networks for data augmentation, quantum entanglement and correlation analysis for feature detection, and quantum-enhanced image reconstruction for noisy data, all performing better than traditional techniques across a wide array of metrics. Specifically, Q-SVM-EFE achieved an accuracy of 96.4% for MRI-based brain tumor detection, compared to 90.2%, 93.0%, and 91.5% obtained using the conventional approaches considered here. The substantial improvement in accuracy may be due to quantum feature extraction, which maps classical image data into high-dimensional quantum states, then retaining important patterns and features. QGAN-DA generated synthesized lung CT scans with a structural similarity index of 0.93, very close to the similarity index of real images measured with an SSIM of 0.95, and their diversity score was 0.88. The high similarity and good diversity of the synthetic images guarantee in this respect that the augmented datasets raise the bar in training deep learning models. It manifests a sensitivity of 97.1% and specificity of 96.0% in detecting microcalcifications in mammograms, which is very high compared with the 90.0%, 92.0%, and 91.0% of the competing methods. High sensitivity and specificity could be attributed to the manner in which this method is capable of leveraging quantum entanglement in detailed correlation analysis. QIR-ND gave a peak signal-to-noise ratio of 42.0 dB with regards to the reconstructed noisy images in CT, with a time to reconstruction of 3.2 seconds and clarity of 0.94. This is distinctly superior to traditional methods, in which PSNR was at best 33.0 dB, 36.2 dB, and 34.5 dB with much longer times to reconstruction. This integration of quantum noise reduction and quantum-enhanced reconstruction techniques enabled QIR-ND to realize efficient operations for high-quality images. In general, the proposed quantum-enhanced approaches ensure leading accuracy, quality, sensitivity, and processing speed for various biomedical image datasets. These results foreshadow the high potential of quantum computing in medical imaging by opening the way toward more correct, effective, and reliable diagnostic tools.

A. FUTURE SCOPE

These promising results from this study open many avenues for future research and development in quantum-enhanced biomedical image processing. First, integration of quantum algorithms with more diverse and larger-scale biomedical datasets could further validate the robustness and scalability of the proposed methods. Additional quantum machine learning techniques, like quantum neural networks and quantum auto-encoders, might bring further improvements in feature extraction and data augmentation capabilities. Besides, hybrid quantum-classical frameworks can be developed, allowing for the power of these two paradigms to be harnessed for more efficient processing and tacking of the current limitations of quantum hardware. Another interesting perspective would be checking how different quantum noise models influence the reliability and accuracy of quantum-enhanced methods in practical applications and applying corresponding error correction techniques. Future work could therefore be focused on real-time implementation and clinical trials for the assessment of practical viability in a healthcare setting. This sort of collaboration would also need medical professionals and researchers to fine-tune the algorithms based on clinical feedback, in order to ascertain that the methods meet the exacting requirements of medical image tasks. In the final analysis, these quantum-boosted methods will continue to improve in performance as quantum computing hardware evolves with increased qubits and fewer errors. Improving in a continuous fashion, quantum technologies likely will expand the scope of applications within biomedical imaging and beyond, leading to new breakthroughs in the medical diagnostics and treatment planning process.

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LALITHA KUMARI PAPPALA received the B.Tech. degree (Hons.) in computer science and engineering and the M.Tech. degree (Hons.) in CSE from JNTUK, India, and the Ph.D. degree from NIT Warangal, Telangana. She is currently an Assistant Professor (Senior Grade 2) with the School of Computer Science and Engineering, VIT-AP University, and has about 18 years of teaching experience. She has published more than 25 publications in refereed international journals and conferences. She has received several best paper awards for her research papers at various international conferences. Her research interests include machine learning, deep learning, and image processing.

SAI BABU VEESAM is currently pursuing the Ph.D. degree with VIT-AP University, Amaravati, India. He has a robust background in IT and academia, with more than seven years of experience in software development and four years as an Assistant Professor. He has contributed significantly to the field through publications in esteemed international journals, conferences, and book chapters. His research interests include computer vision, machine learning, quantum computing, and artificial intelligence.

KANNAIAH CHATTU received the Ph.D. degree from VIT AP University, Amaravati, Andhra Pradesh. He is currently an Assistant Professor with the Department of Computer Science and Engineering (AIML), Malla Reddy College of Engineering and Technology, Maisammaguda, Bhadurpalle, Hyderabad, Telangana, India. He had ten years of teaching experience and three years of research experience. He has published research articles in national and international scientific journals indexed in the Scopus database. His research interests include natural language processing, deep learning, and machine learning.

JANAPATI VENKATA KRISHNA received the B.Tech. degree in CSE from ANU, the M.Tech. degree in CST from AU, and the Ph.D. degree in CSE from JNTU Hyderabad. Currently, he is an Associate Professor with the Unit Head of IoT and Robotics and AI and AR&VR Departments, Institute of Engineering and Technology, Srinivas University, Mangaluru, India. He qualified in GATE-2002. He also qualified in FET-2010 conducted by JNTUH and was ratified as an Assistant and an Associate Professor by JNTUH, in 2011. He has four years of industry (TCS) and 18 years of teaching experience. He has published around 50 international journals with Google Scholar citations. He has also presented more than 20 papers at national and international conferences. He authored two books with an ISBN and added chapters for more than five books. He also has two patents registered. His research interests include cloud computing, the IoT, and big data analytics.

JYOSTNA DEVI BODAPATI (Senior Member, IEEE) received the M.Tech. and Ph.D. degrees in computer science and engineering. Currently, she is an Associate Professor with the Department of Advanced Computer Science and Engineering, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Andhra Pradesh, India. She was a Research Assistant with IIT Madras, from 2013 to 2016. She is recognized among the top 2% of researchers worldwide, according to the 2024 report by Stanford University. With extensive expertise in teaching and research, she has published more than 50 papers in reputed international journals and conferences. Her research interests include deep learning and computer vision, with a focus on sequential data analysis.

B. TARAKESWARA RAO received the M.Tech. and Ph.D. degrees from Acharya Nagarjuna University, Guntur, Andhra Pradesh, in 2007 and 2012, respectively. He is currently a Professor with the Department of Computer Science and Engineering, Kallam Haranadhareddy Institute of Technology, Guntur. He published more than 50 research articles in various international journals and presented more than ten research papers at various national and international conferences. He published ten book chapters in various books. He has 21 years of experience in teaching and research. His research interests include artificial intelligence, data science, machine learning, big data, and deep learning.